MIL-W-81116B
15 November 1972
Superseding
MIL-T-81116A(ASG)
10 May 1965

MILITARY SPECIFICATION

WEBBING, TEXTILE, POLYAMIDE, HIGH TEMPERATURE RESISTANT, LOOP

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for a high temperature resistant loop webbing used as a lacing tape in flight clothing.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATION

Military

MIL-W-43334	Webbing and Tape, Textile, Packaging and Packing of

STANDARD

Federal

FED-STD-191	Textile Test Methods

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

FSC 8305

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Rules and Regulations Under the Textile Fiber Products Identification Act

(Application for copies should be addressed to the Federal Trade Commission, Washington, D.C. 20580.)

3. REQUIREMENTS

* 3.1 Standard sample - The webbing shall match the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

* 3.2 Materials - The webbing shall be produced from a high temperature aromatic polyamide yarn. The yarn shall not char at a temperature less than 675°F (357°C).

3.2.1 Loop cord - The loop cord used in the manufacture of the lacing loop webbing shall be constructed in such a manner and with such yarn tension that the core yarns are held firmly in place to prevent puckering of the core yarns when released from stress. The cord shall conform to the requirements specified in Table I.

3.3 Webbing - The finished webbing shall conform to the requirements specified in Table II.

3.3.1 Weave - The finished loop webbing shall be woven in accordance with the weave diagram shown in Figure 1.

3.3.2 Length relationship between loop cord and finished webbing - The finished webbing shall contain a length of 45 ± 1 inch of cord for each yard of webbing in order to produce the desired loop opening.

* 3.3.3 Color - The color of the webbing shall be as specified by the procuring activity. The color shall be obtained by the use of melt spun solution dyed fibers.

* 3.3.4 Matching - The color shall match the standard sample under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

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TABLE I. CORD PHYSICAL PROPERTIES

CHARACTERISTICS	REQUIREMENTS		
	SLEEVE	CORE	CORD
Yarn size, denier	200	200	
Yarn ply, minimum	3	3	
Yarn twist, turns per inch, minimum			
Spin	8S	8S	
Ply	6Z	6Z	
Number of carriers	20	-	
Ends per carrier	1	-	
Number of ends	20	6	
Picks per inch	26-28	-	
Weave	Plain	-	
Breaking strength, pounds, minimum	-	-	150
Yards per pound, minimum	-	-	200

TABLE II. WEBBING PHYSICAL PROPERTIES

CHARACTERISTICS	REQUIREMENTS
Yarn size, denier	200
Yarn ply	
Warp	Single
Filling	3
Yarn count, minimum	
Face warp	38
Back warp	37
Cord warp	1
Total warp	76
Filling per inch	60
Width, inch $\pm$ 1/32	
Webbing and cord woven together	5/16
Webbing and cord loop	7/16
Yards per pound, maximum	67
Breaking strength, pounds, minimum	175

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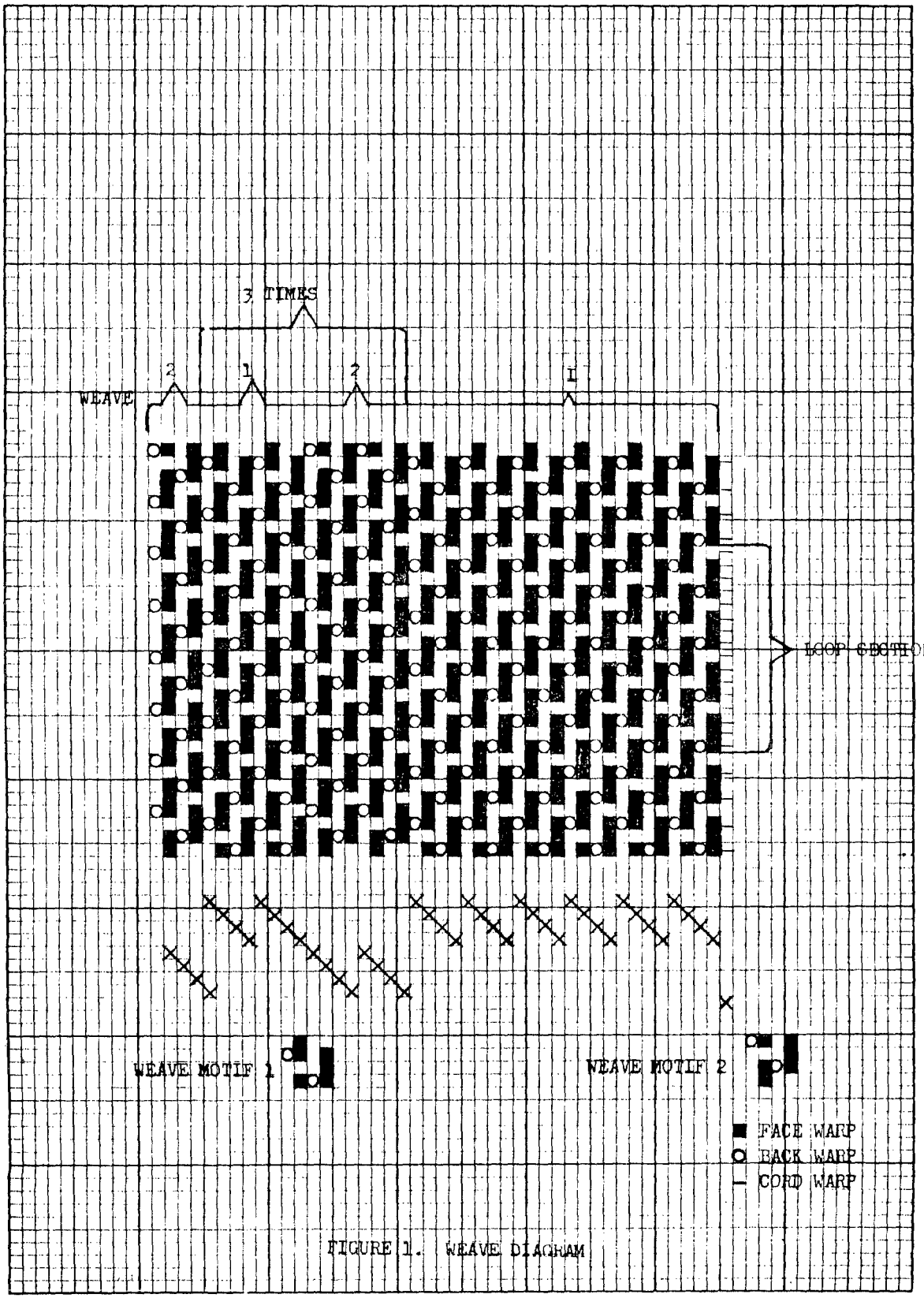


FIGURE 1. WEAVE DIAGRAM

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* 3.4 pH - The pH value of the water extract of the finished webbing shall be no less than 5.0 nor more than 8.0 when tested as specified in 4.3.

3.5 Length of spool - Unless otherwise specified, each spool of webbing shall contain 50 ± 2 yards. No spool shall contain more than 2 pieces, and no piece shall be less than 10 yards in length.

* 3.6 Identification tickets - Each spool of webbing shall be labeled, ticketed, or invoiced for fiber content in accordance with the Textile Products Identification Act.

* 3.7 Workmanship - The webbing shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the acceptable quality levels specified herein.

*4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Certificate of compliance - Where certificates of compliance are submitted in accordance with 4.3, they shall contain verifiable actual test inspection data. The Government reserves the right to inspect and test webbing to verify the validity of the certification.

* 4.2 Inspection - Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

* 4.2.1 Examination of the end item - Defects found during the examination shall be classified in accordance with 4.2.1.1, 4.2.1.2, and 4.2.1.3.

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* 4.2.1.1 Yard-by-yard examination - The sample of webbing shall be examined on both sides and the defects shall be classified in accordance with Table III. The defects found shall be counted regardless of their proximity to each other except where two or more defects represent a single local condition of the webbing in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs. The sample unit for this examination shall be 1 linear yard. The sample size for this examination shall be selected in accordance with MIL-STD-105, Inspection Level II, with an acceptable quality level of 2.5 major defects and 10 total defects per 100 units (yards). The required sample yardage shall comprise approximately an equal number of yards out of each selected spool. The number of spools from which the sample is to be selected shall be in accordance with Table IV.

* 4.2.1.2 Overall examination - Each defect listed below shall be counted no more than once in each spool examined. The sample unit for this examination shall be one spool. The sample size and acceptance number shall be as shown in Table IV. The lot size shall be expressed in units of 1 linear yard.

*Defects

Off shade, not within established tolerances.
Cloudy, mottled or streaky throughout.
Poorly constructed, not firmly and tightly woven.
Poor color penetration.
Objectional odor.
Overall uncleanness.
Not labeled in accordance with Textile Fiber Products Identification Acts.

4.2.1.3 Examination of length -

* 4.2.1.3.1 Individual spool - Each spool in the sample shall be examined for the following defects. The sample unit for this examination shall be one spool. The sample size and acceptance number shall be as shown in Table IV. The lot size shall be expressed in units of 1 linear yard each.

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TABLE III. CLASSIFICATION OF DEFECTS

DEFECT	DEFINITION	CLASSIFICATION MAJOR	CLASSIFICATION MINOR
Abrasion mark	Resulting in rupture of individual yarns or plies, distortion in the orientation of threads, dimensional distortion, areas noticeably thinner than adjoining unaffected areas, or in nap sufficient to obscure the identity of any yarns.	X	
Broken and missing ends or picks	Two or more contiguous regardless of extent.	X	
	Single		X
Coarse filling bar	Resulting in noticeably increased stiffness or thickness of webbing.	X	
Crease or wrinkle, edge folded or rolled	Hard, embedded		X
Cut, hole, or tear	Any cut, hole, or tear	X	
Drop-ply	Two or more drop-ply end in the same length.	X	
	Single drop-ply end in the same length.		X
Edge cut, torn or frayed	Any cut, torn, or frayed edge.	X	
Edge beaded or corded	Noticeable increase in thickness, or misformed edge.		X
Edge loopy	Forming clearly visible filling loops, or edge tied loosely to body of webbing for 2 linear inches or more 1/	X	
Edge loose (slack)	Resulting in waviness, distortion in orientation of filling, or looseness along edge.	X	

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TABLE III. (Cont'd)

DEFECT	DEFINITION	CLASSIFICATION MAJOR	CLASSIFICATION MINOR
Edge tight	Resulting in noticeable tension along edge, or pucker, waviness, bagginess, or slackness that cannot be flattened by manual pressure.	X	
Scalloped edge	Any noticeable indentation of edge.		X
Fine or light filling bar, light place	Clearly noticeable	X	
Floats or skips	Any multiple floats or skips.	X	
	Single float or skip more than 1/4 inch in length in warp or full width of the tape in the filling direction.	X	
	Single, floating for 1/4 inch or less in warp direction, or up to full width of the webbing in the filling direction.		X
Hitch-back crack, open place	Clearly noticeable opening between adjoining picks or warpwise tension area over part of the width, resulting in noticeable light and heavy places.		X
Jerked-in filling, slough-off, slug	More than twice the thickness of the normal yarn.		X
Kinks	More than three in any 9 linear inches.	X	
Knots	More than one knot in any 9 linear inches.	X	
	Single knot with untrimmed ends extending from surface of tape.		X
Mispick, double pick	Two or more contiguous	X	
	Single		X

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TABLE III. (Cont'd)

DEFECT	DEFINITION	CLASSIFICATION MAJOR	CLASSIFICATION MINOR
Slack end	Two or more in same length, jerked in between picks, or forming clearly noticeable loops on surface of tape.	X	
Smash	Any smash.	X	
Spot, stain, or streak	Any clearly noticeable 1/ dirt, rust, grease, oil spots, stain, or streak.		X
Tight end	Clearly noticeable 1/	X	
Tight pick or tight filling	Resulting in rolling of tape (see also "edge scalloped").	X	
Weak, soft, or tender spot	Any weak, soft, or tender spot.	X	
Wrong draw	Extending for more than 9 inches.		X
Width	Beyond tolerance.		X

1/ Clearly visible at normal distance of 3 feet.

* TABLE IV

LOT SIZE IN YARDS	SAMPLE SIZE IN SPOOLS	MAXIMUM NUMBER OF DEFECTS ACCEPTABLE IN SAMPLE 1/
up to 1200	3	0
1201 up to and including 3200	5	0
3201 up to and including 10,000	8	0
10,001 up to and including 35,000	13	0
35,001 up to and including 150,000	20	1
150,001 and over	32	2

1/ If a lot contains fewer than 3 spools, each roll in the lot shall be examined.

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* Defect

Gross length less than specified minimum length or more than specified maximum length.

Gross length more than 2 yards less than gross length marked on piece ticket.

Any piece less than the allowable minimum length of piece.

Any spool containing more than the allowable number of pieces permitted for the applicable type of webbing.

* 4.2.1.3.2 Total yardage in sample - The lot shall be unacceptable if the total of the actual gross lengths of spool in the sample is less than the total of the gross lengths marked on the spool tickets.

* 4.2.2 Examination of preparation for delivery requirements - An examination shall be made in accordance with the provisions of MIL-W-43334 to determine that packaging, packing, and marking comply with the section 5 requirements.

* 4.3 Testing of the end item - The methods of testing specified in FED-STD-191, wherever applicable, and as listed in Table V shall be followed. When the data in the "number of determinations" and "results reported as" columns are not specified in the table, they shall be as required by the referenced test method. The physical and chemical values specified in section 3 apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test methods. All test reports shall contain the individual values utilized in expressing the final result. The sample size shall be as follows:

Lot size (yards)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

The lot shall be unacceptable if one or more units fail to meet any requirement specified. The lot shall be expressed in units of 1 linear yard. The sample unit for testing shall be 10 yards.

* 5. PREPARATION

* 5.1 Packaging - Packaging shall be level A, B, or C as specified (see 6.2).

* 5.1.1 Levels A, B, and C - Webbing, put up as specified, shall be packaged in accordance with the applicable requirements of MIL-W-43334.

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TABLE V. TEST METHOD

CHARACTERISTICS	REQUIREMENT REFERENCE	TEST METHOD
Identification	3.2	1/
Char point	3.2	1/
Denier	Tables I, II	4021 1/
Ply	Tables I, II	visual
Twist	Table I	4054
Ends per carrier	Table I	visual
Number of ends	Table I	visual
Picks per inch	Table I	visual
Yarn count	Table II	5050
Breaking strength	Tables I, II	4102
Yards per pound	Tables I, II	5040
Number of carriers	Table I	visual
Weave	Figure I, Table I	visual
Melt spun dyed	3.3.3	1/
pH	3.4	2811

1/ Unless otherwise stated, a certificate of compliance shall be submitted and will be acceptable for the stated requirements.

* 5.2 Packing - Packing shall be level A, B, or C as specified (see 6.2).

* 5.2.1 Levels A, B, and C - The webbing shall be packed in accordance with the applicable requirements of MIL-W-43334.

* 5.3 Marking - In addition to any special marking required by the contract or order, shipments shall be marked in accordance with MIL-W-43334.

6. NOTES

6.1 Intended use - The webbing is intended for use as a lacing in the anti G coveralls, Mark 2A, and related clothing.

6.2 Ordering data - Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Length of spool required (if other than specified in 3.5).

(c) Color of webbing required (see 3.3.3).

(d) Selection of applicable levels of packaging and packing (see 5.1).

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6.3 Standard sample - For access to standard sample, address the procuring office issuing the invitation for bids.

6.4 The margins of the specification are marked with an asterisk to indicate where changes (additions, modifications, corrections) from the previous issues were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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DOCUMENT IDENTIFIER AND TITLE
MIL-W-81116B WEBBING, TEXTILE, POLYAMIDE, HIGH TEMPERATURE RESISTANT, LOOP

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO (If "Yes", in what way?)

4. REMARKS

SUBMITTED BY (Printed or typed name and address - Optional)

TELEPHONE NO.

DATE

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